

3,500 ft Rocket — Electronics & Motor Sourcing

Reference sheet for Step 1 (OpenRocket modeling) and Step 2 (ordering) of the project plan.

Flight Computer Electronics

Generic/clone parts — cheap and well-documented, matching the frugal DIY approach in the project plan.

Function	Part	Why This One / Notes	Est. Cost
Microcontroller	Arduino Nano (clone)	Small, breadboard-friendly, huge library support	\$3–6
Altitude	BMP388 barometric sensor	Better resolution & faster sampling than BMP280	\$3–7
Accel / gyro	MPU6050	Cheapest common 6-axis IMU, well-supported libraries	\$2–5
Position	NEO-6M GPS module	See COCOM limit warning below before buying	\$8–15
Storage	MicroSD card module (SPI) + microSD card	Backup log if radio link drops out mid-flight	\$3–6
Radio	2× RFM95W LoRa module (915 MHz)	One for rocket, one for ground station	\$8–15 each
Power	1S LiPo (500–900mAh) + buck/boost regulator	Match regulator output to your sensors' voltage	\$6–12

GPS Warning — Read Before Buying

Cheap GPS modules like the NEO-6M often ship with a built-in altitude/speed limit (commonly around 1,000–1,500 m altitude and/or 500 m/s velocity), due to COCOM export restrictions meant to prevent missile-guidance use.

3,500 ft is about 1,067 m — right at that edge — and G-motor rockets can reach speeds where the velocity limit matters too. Before purchasing, check the listing for “COCOM limits” or look specifically for a module advertised as unlocked / high-altitude. Otherwise the GPS may simply stop reporting mid-flight.

Motor Sourcing

No NAR/Tripoli certification is required for G-class motors — they're sold over the counter as standard hobby items.

Source	Notes
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Apogee Rockets	Carries 29mm AeroTech / Cesaroni G motors, 100–150 Ns range. Also body tubes, nose cones, fin stock.
Wildman Rocketry	Good for price comparison on the same motor classes and airframe parts.
Local hobby shop	Worth a call — shops that stock rocketry (not just RC cars) may carry Estes/AeroTech G motors, saving on shipping.
ThrustCurve.org	Not a vendor — use this to pull exact thrust-curve data for whichever motor you buy, to import into OpenRocket.

Motor Selection Guidance

- Target 29mm motor mount
- Look for AeroTech or Cesaroni G-series motors in the 100–150 Ns total impulse range (e.g. AeroTech G74, G77)
- Wait to purchase the motor until close to your actual launch date — no reason to store propellant for weeks

Vendor links:

[Apogee Rockets](#)

[Wildman Rocketry](#)

[ThrustCurve.org](#)